LESSON PLAN

AFCS (22)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	08
Total No. of Lesson Plans	10	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Long range guidance: Inertial navigation, Hybrid inertial system and air flying missile guidance.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Long range guidance: Inertial navigation, Hybrid inertial system and air flying missile guidance.
5. Specific Objectives	22.1.1.1Trainee will be able to understand the basic of guided weapon systems and types of targets.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of Inertial Navigation System components and Missile Guidance modes using L & Ds and IB.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary types of long-range guidance systems?
(b) Define the basic principle of inertial navigation.
(c) List the common types of targets for guided weapons.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic principles of inertial navigation • Components of inertial navigation system • Error accumulation in inertial systems • Hybrid inertial system architecture • Integration of GPS with INS	L & Ds	PPT, IB	15 min
(b)	• Air-to-air missile guidance modes • Air-to-surface missile guidance modes • Terminal guidance techniques • Missile seeker types and operation • Guidance law selection criteria	L & Ds	PPT, PDF	20 min
(c)	• Comparison of inertial and hybrid systems • Advantages of hybrid inertial guidance • Limitations of pure inertial navigation • Future trends in missile guidance • Operational considerations for aircrew	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how a hybrid inertial system corrects drift errors.
(b) Differentiate between semi-active and active radar homing in missiles.
(c) Describe the role of the inertial reference unit in a missile.

12. Home Assignment and Follow Up	Prepare a brief note on the specific guidance units and reference units discussed in the next lesson plan (LP 09) and how they interface with the inertial system.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Sensor units for guidance systems computer units of guidance system, reference units of guidance system.	Refer to the master syllabus reference list	LP 09

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min